NAFLAN MOHAMED

AI Engineer | Applied AI & LLM Systems | Python Backend Developer

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PROFESSIONAL SUMMARY

Computer Science graduate and AI/ML Engineer with hands-on experience across three software and AI internships, building LLM applications, Retrieval-Augmented Generation (RAG) systems, AI agents, recommendation systems, and Python backend services. Experienced with Python, FastAPI, LangChain, LangGraph, PostgreSQL, vector databases, and asynchronous APIs. Built end-to-end AI applications covering document processing, embeddings, semantic retrieval, tool calling, agent orchestration, backend APIs, persistence, and deployment.

TECHNICAL SKILLS

Programming: Python, SQL, Java, JavaScript, C++

Generative AI & LLMs: LLM Applications, RAG, LangChain, LangGraph, Gemini API, AI Agents, Agentic Workflows, Tool/Function Calling, Prompt Engineering, Model Context Protocol (MCP)

Machine Learning & NLP: Scikit-learn, TensorFlow, Classification, Regression, Clustering, Recommender Systems, Neural Networks, Sentence Transformers, Embeddings, Semantic Search, Vector Search, HNSW Indexing

Backend & APIs: FastAPI, Django REST Framework, REST APIs, Asynchronous Programming, Authentication, Webhooks

Databases & Retrieval: PostgreSQL, pgvector, ChromaDB, Redis, Supabase, MongoDB

Frontend / Cloud: React.js, Next.js, Node.js, HTML, CSS, Docker, Kubernetes, Google Cloud Platform (GCP)

Computer Science: Object-Oriented Programming (OOP), Data Structures & Algorithms

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern | NETSOL Technologies | Lahore, Pakistan **Jul 2026 - Sep 2026**

- Developed and integrated 5+ LLM-powered capabilities using Python, FastAPI, LangChain, LangGraph, and Gemini, supporting RAG, web search, email, calendar, and tool-calling workflows.
- Engineered an end-to-end RAG pipeline for 50+ research documents and 2,000+ text chunks, covering document parsing, chunking, 768-dimensional embeddings, vector retrieval, context construction, and LLM response generation.
- Evaluated retrieval and prompt behavior across 100+ test queries to improve retrieval relevance and reduce irrelevant context supplied to the LLM.
- Built stateful LangGraph agent workflows integrating 4+ tools with dynamic routing between document retrieval, web search, calendar management, and email operations.
- Implemented asynchronous processing, error handling, and fallback mechanisms across LLM and external API workflows to improve reliability during tool and API failures.

MERN Stack Developer Intern | Fluxxion **Jun 2025 - Aug 2025**

- Refactored React component architecture using memoization and lazy loading, improving Lighthouse performance scores by 40% and reducing reported UI defects by 30%.
- Designed and delivered a modular Daily Goals Tracker feature with test coverage and integration into the core application.
- Optimized MongoDB aggregation pipelines and introduced field-level indexing for high-traffic REST API endpoints to improve backend query and response performance.

Web Developer Intern | AICP **Mar 2024 - Jun 2024**

- Developed and maintained 5+ web applications using HTML, CSS, and JavaScript, including client-side validation and responsive interfaces.
- Refactored frontend code across multiple projects to improve mobile responsiveness and eliminate major layout issues across tested screen sizes.
- Integrated third-party APIs and dynamic content feeds into web applications to extend product functionality.

SELECTED PROJECTS

AI ChatAgent Workspace

Tech: Python, FastAPI, LangGraph, Gemini, RAG, MCP, React, PostgreSQL, Supabase, Redis | github.com/naflanmohamed/ChatAgent | https://chattingagent.vercel.app/

- Built a full-stack agentic AI workspace integrating 5+ capabilities including web search, RAG document retrieval, URL analysis, email, and calendar automation.
- Designed stateful LangGraph workflows with conditional execution, dynamic routing, tool calling, and persistent state for multi-step AI tasks.
- Integrated MCP and asynchronous FastAPI services with PostgreSQL/Supabase persistence, Redis state management, authentication, external tools, and fault-tolerant execution.

Yaloo - AI-Powered Cultural Tourism Platform | Final Year Project

Tech: Flutter, Django REST Framework, PostgreSQL, Supabase, RAG, pgvector, Sentence Transformers, Gemini, Redis

- Architected an AI-powered tourism platform connecting tourists with verified guides and homestay hosts, including booking, availability, reviews, authentication, and real-time interactions.
- Developed a recommendation system across three domains using Sentence Transformers embeddings, pgvector similarity search, HNSW indexing, and hybrid scoring.
- Built an intent-routed Gemini RAG assistant with query validation, history-aware retrieval, context injection, and multi-level fallback handling.

DocChat - AI Document Question-Answering Platform

Tech: Next.js, FastAPI, Python, RAG, LangChain, Gemini API, ChromaDB | docchats.vercel.app

- Built and deployed a document-grounded conversational AI application supporting PDF, DOCX, and TXT files with source-aware question answering.
- Developed the complete RAG pipeline covering parsing, recursive chunking, embeddings, ChromaDB vector storage, semantic retrieval, context construction, and Gemini-powered generation.
- Evaluated retrieval performance across 50+ test queries and 1,500+ document chunks using Precision@K, Recall@K, and response latency.

EDUCATION

Bachelor of Computer Science (Hons). Dec 2022 - Sep 2026

Government College University (GCU), Lahore | Allama Iqbal Scholarship Recipient

CERTIFICATIONS

- Machine Learning Specialization - IBM
- ChatGPT Prompt Engineering for Developers - DeepLearning.AI
- Full Stack Developer: Front-End & Back-End from Scratch Specialization - Meta